Drawing Travel Graphs

Exam code: 9MA0

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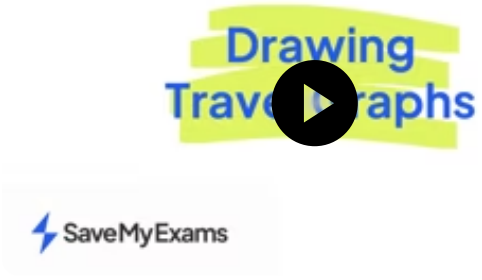

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Drawing travel graphs

How do I draw a displacement-time graph or a velocity-time graph?

- You may be asked to draw a **displacement-time** graph or a **velocity-time** graph
- You will be given information about different sections of a journey and must draw each section and **label** the **points** on the axes
- Remember that time will always be on the x-axis
- Always remember to include **units** when you label the axis
- Use appropriate facts about the **gradient** and **area** under the graph to work backwards and find relevant or missing information
  - The **gradient** of a **displacement-time** graph is the **velocity** and the **gradient** of a **velocity-time** graph is the **acceleration**
  - The **area** between a **velocity-time** graph and the x-axis is the **displacement**

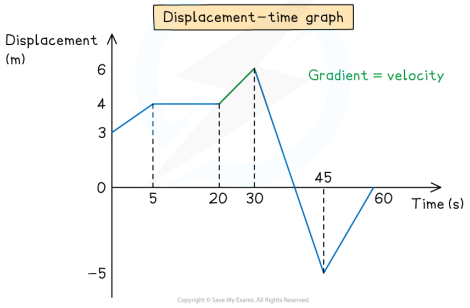
Help me understand

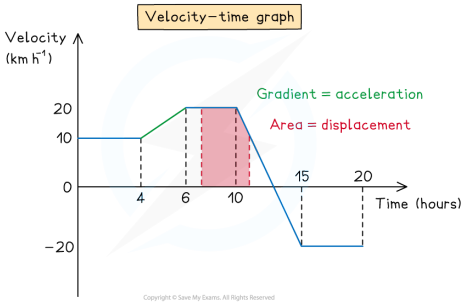
Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

Step 1: Draw in the lines to complete the graph



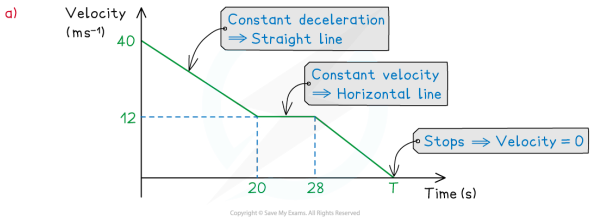


Worked Example

The driver of a train, travelling at 40 m s^{-1} on a straight horizontal track, is informed of debris on the track 0.74 km ahead and immediately applies the brakes.
The train decelerates uniformly for 20 seconds to a speed of 12 m s^{-1}
The train then maintains this speed for another 8 seconds.
The driver reapplies the brakes to slow down the train at a constant rate and bring it to a complete stop 10 m before the debris.

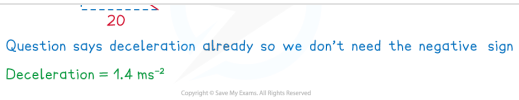
(a) Sketch a velocity-time graph to show the motion of the train.

Answer:



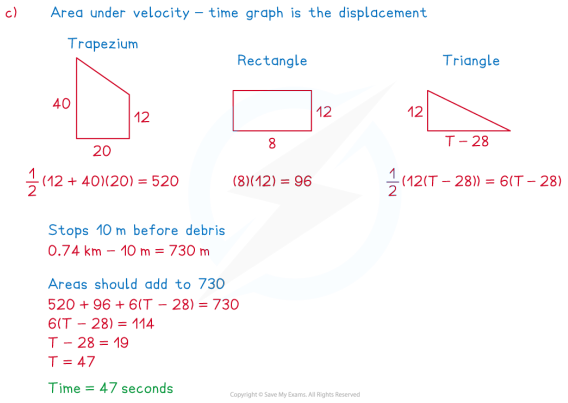
(b) Find the deceleration of the train from the moment the brakes were first applied to the moment its speed first reached 12 m s^{-1}.

Answer:



(c) Calculate the total time from the moment the brakes were first applied to the moment the train came to rest.

Answer:



Examiner Tips and Tricks

- As in the worked example, examiners can use other words, such as uniformly, to mean constant.
- Be careful to spot if you are working with a displacement-time graph or a velocity-time graph.
- Remember that displacement and velocity can be negative whereas distance and speed can **not** be negative.
- Take care when a velocity-time graph is below the x-axis, if it has a negative gradient then it is speeding up and moving backwards. If it has a positive gradient below the x-axis then it is still moving backwards but it is slowing down.
- Think about the units when calculating displacement, velocity and acceleration.

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